

Machine Learning Interview Questions and Answers

1. What is Machine Learning?

Machine Learning is a subset of Artificial Intelligence that provides systems the ability to learn and improve from experience without being explicitly programmed.

2. What are the types of Machine Learning?

Supervised Learning, Unsupervised Learning, Semi-supervised Learning, and Reinforcement Learning.

3. What is Supervised Learning?

Supervised Learning involves training a model on a labeled dataset, meaning each training example is paired with an output label.

4. What is Unsupervised Learning?

Unsupervised Learning deals with unlabeled data and the model tries to identify patterns or groupings in the data.

5. What is Reinforcement Learning?

Reinforcement Learning is where an agent learns to make decisions by performing actions and receiving rewards or penalties.

6. What is Overfitting?

Overfitting occurs when a model learns the training data too well, including noise and outliers, and performs poorly on new data.

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7. What is Underfitting?

Underfitting occurs when a model is too simple and fails to capture the underlying pattern of the data.

8. What is a Training Set and a Test Set?

The training set is used to train the model, while the test set is used to evaluate the model's performance.

9. What is Cross-validation?

Cross-validation is a technique for evaluating machine learning models by partitioning the data into complementary subsets, training on one subset and validating on the other.

10. What is a Confusion Matrix?

A confusion matrix is a summary of prediction results on a classification problem, showing the number of correct and incorrect predictions.

11. What is Precision?

Precision is the ratio of true positive predictions to the total predicted positives.

12. What is Recall?

Recall is the ratio of true positive predictions to all actual positives.

13. What is F1-Score?

F1-Score is the harmonic mean of precision and recall and is a better measure than accuracy in cases of imbalanced classes.

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14. What is Accuracy?

Accuracy is the ratio of correctly predicted observations to the total observations.

15. What is the Bias-Variance Tradeoff?

Bias is the error due to overly simplistic assumptions. Variance is the error from sensitivity to small fluctuations in the training set. The tradeoff is finding a balance between the two.

16. What is a ROC curve?

ROC (Receiver Operating Characteristic) curve is a graphical plot that illustrates the diagnostic ability of a binary classifier.

17. What is AUC - ROC?

AUC (Area Under the Curve) represents the degree or measure of separability. AUC-ROC curve tells how much the model is capable of distinguishing between classes.

18. What is Gradient Descent?

Gradient Descent is an optimization algorithm used to minimize the cost function in a model by iteratively adjusting the parameters.

19. What are the types of Gradient Descent?

Batch Gradient Descent, Stochastic Gradient Descent, and Mini-batch Gradient Descent.

20. What is a Cost Function?

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A cost function measures how well a machine learning model performs. It represents the difference between the predicted values and the actual values.

21. What is Regularization?

Regularization is a technique used to reduce overfitting by adding a penalty term to the cost function.

22. What are L1 and L2 Regularization?

L1 regularization adds the absolute value of coefficients as penalty; L2 adds the square of coefficients.

23. What is Feature Engineering?

Feature Engineering is the process of selecting, modifying, or creating new features from raw data to improve model performance.

24. What is Dimensionality Reduction?

Dimensionality Reduction reduces the number of random variables under consideration, using techniques like PCA.

25. What is PCA?

PCA (Principal Component Analysis) is a technique used to emphasize variation and bring out strong patterns in a dataset by reducing dimensions.

26. What is Clustering?

Clustering is an unsupervised learning method used to group similar data points into clusters.

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27. What are some clustering algorithms?

K-Means, DBSCAN, Hierarchical Clustering, Mean Shift.

28. What is K-Means Clustering?

K-Means is a clustering algorithm that partitions data into K distinct clusters based on distance to the centroid of a cluster.

29. What is the Elbow Method?

The Elbow Method is used to find the optimal number of clusters by plotting the cost (within-cluster sum of squares) versus number of clusters.

30. What is a Decision Tree?

A Decision Tree is a flowchart-like structure used for classification and regression by splitting the dataset based on feature values.

31. What is Random Forest?

Random Forest is an ensemble of decision trees that improves classification or regression performance by averaging multiple decision trees.

32. What is Ensemble Learning?

Ensemble Learning combines multiple models to produce a better result than any single model, using methods like Bagging and Boosting.

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33. What is Bagging?

Bagging (Bootstrap Aggregating) builds multiple models (usually the same type) from different subsamples and averages the results.

34. What is Boosting?

Boosting builds models sequentially, with each new model trying to correct the errors made by the previous ones.

35. What is XGBoost?

XGBoost is an optimized gradient boosting library designed to be highly efficient, flexible, and portable.

36. What is Naive Bayes?

Naive Bayes is a classification algorithm based on Bayes' Theorem with an assumption of independence among predictors.

37. What is SVM?

Support Vector Machine is a supervised learning algorithm that finds the hyperplane that best separates different classes.

38. What is Kernel Trick in SVM?

Kernel Trick is a method of transforming data into a higher dimension so that it becomes separable using a linear classifier.

39. What is ROC-AUC Curve used for?

To evaluate the performance of a classification model by showing the trade-off between sensitivity and specificity.

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40. What is the Curse of Dimensionality?

The Curse of Dimensionality refers to the problem that arises when the feature space becomes too large, leading to sparse data and reduced model performance.